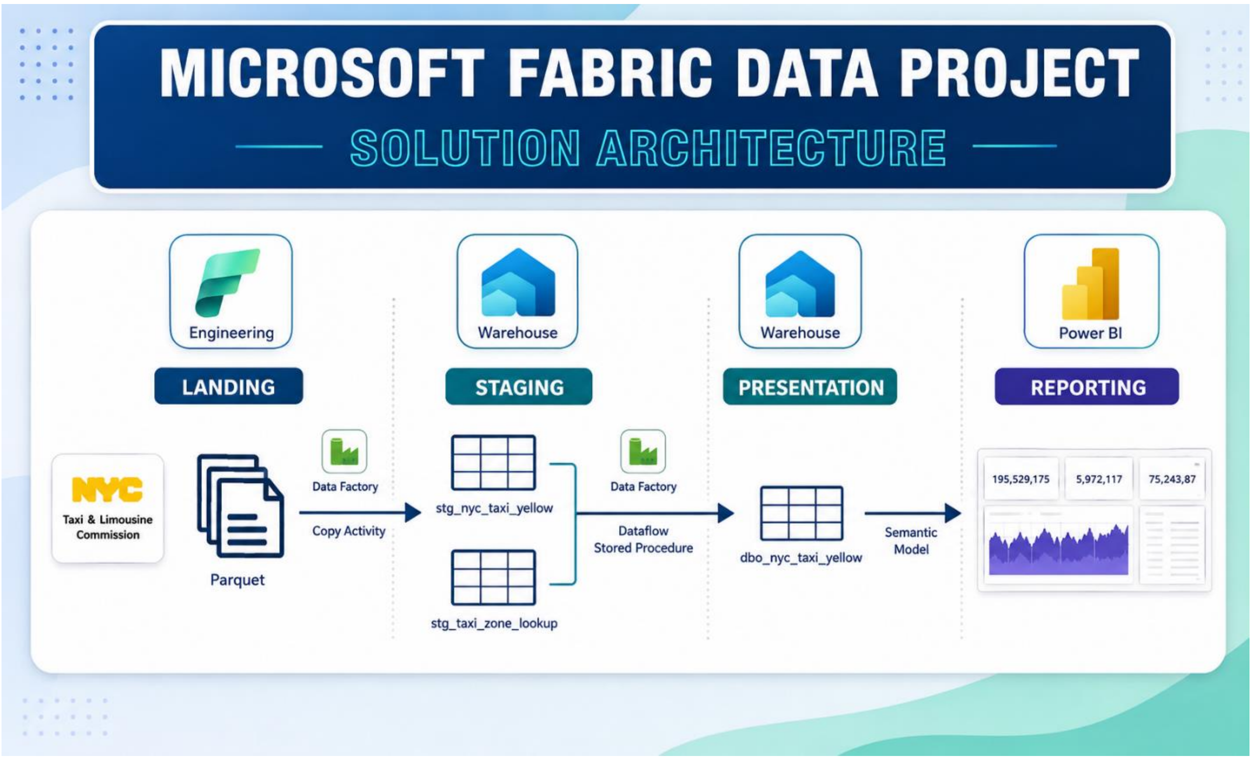
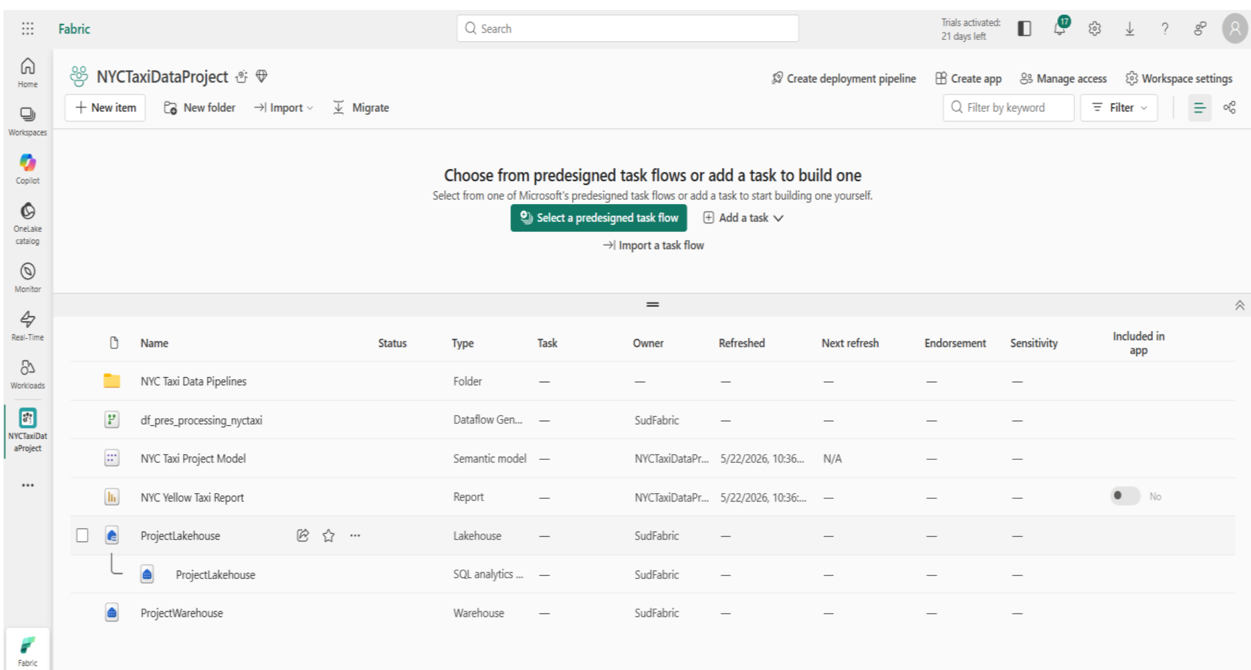
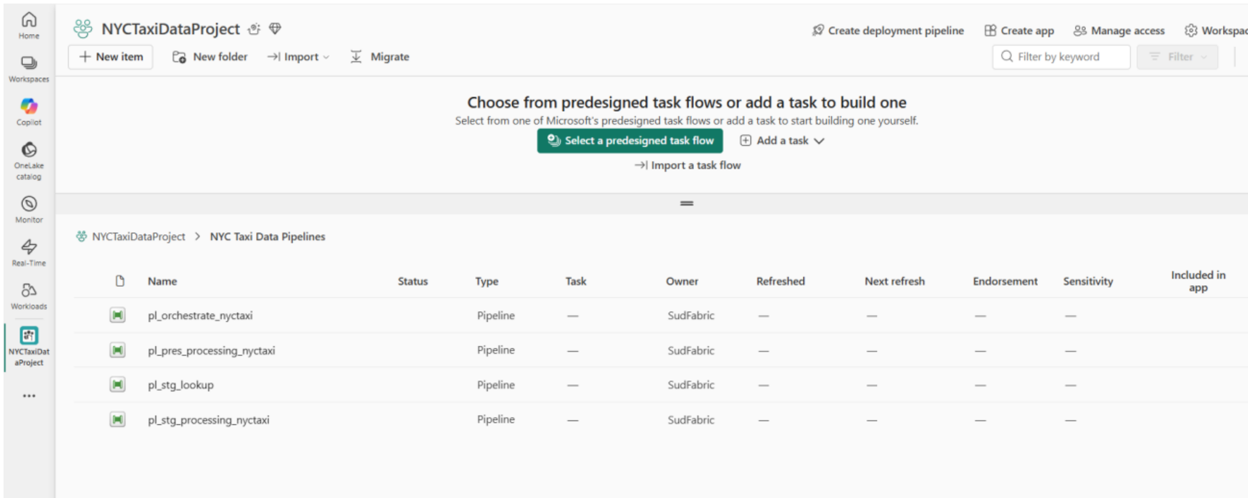
Project Architecture



Created a Fabric Workspace

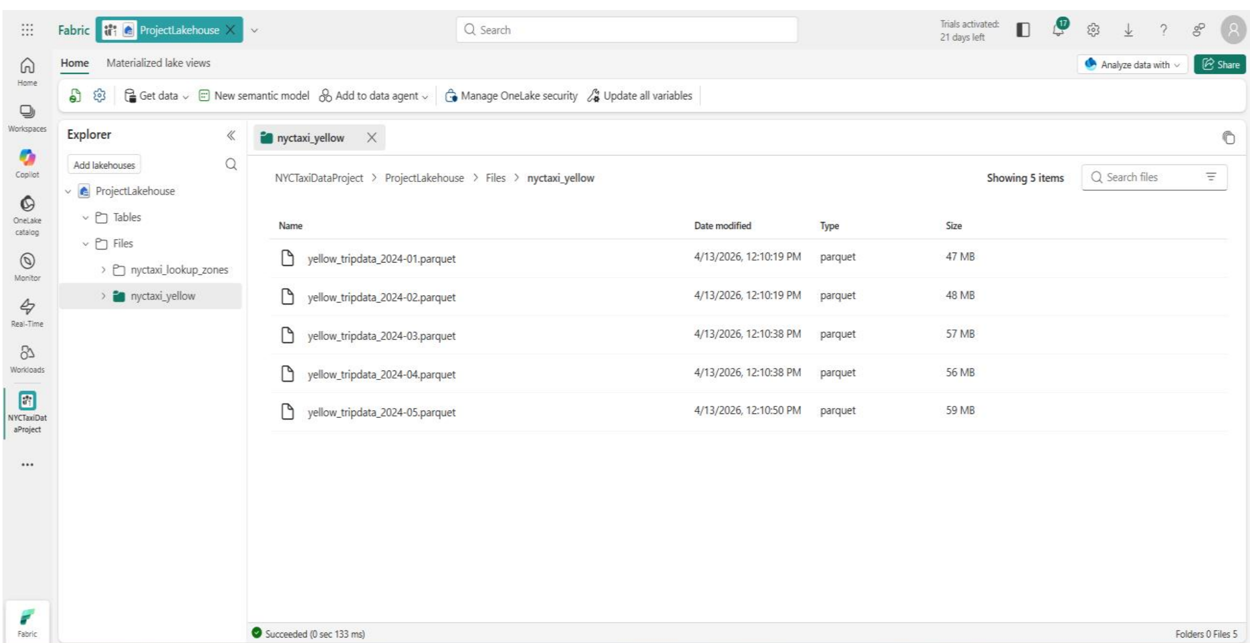


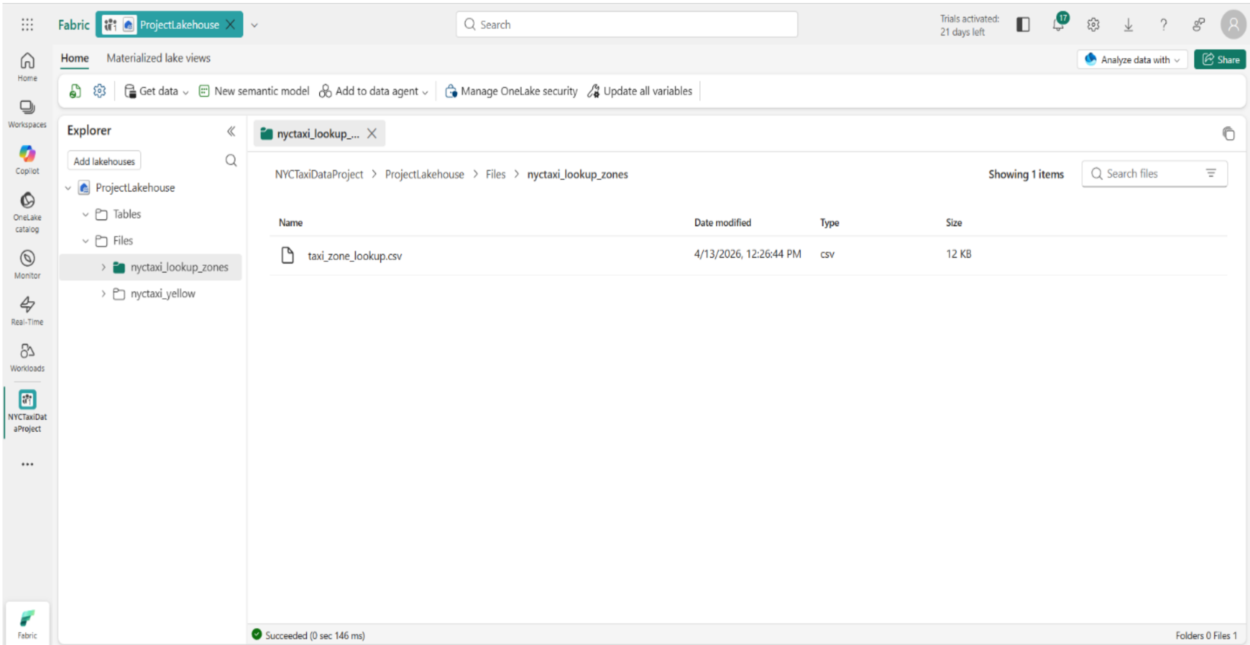
Data Pipelines created in this Workspace



<u>Landing the data in lakehouse</u>

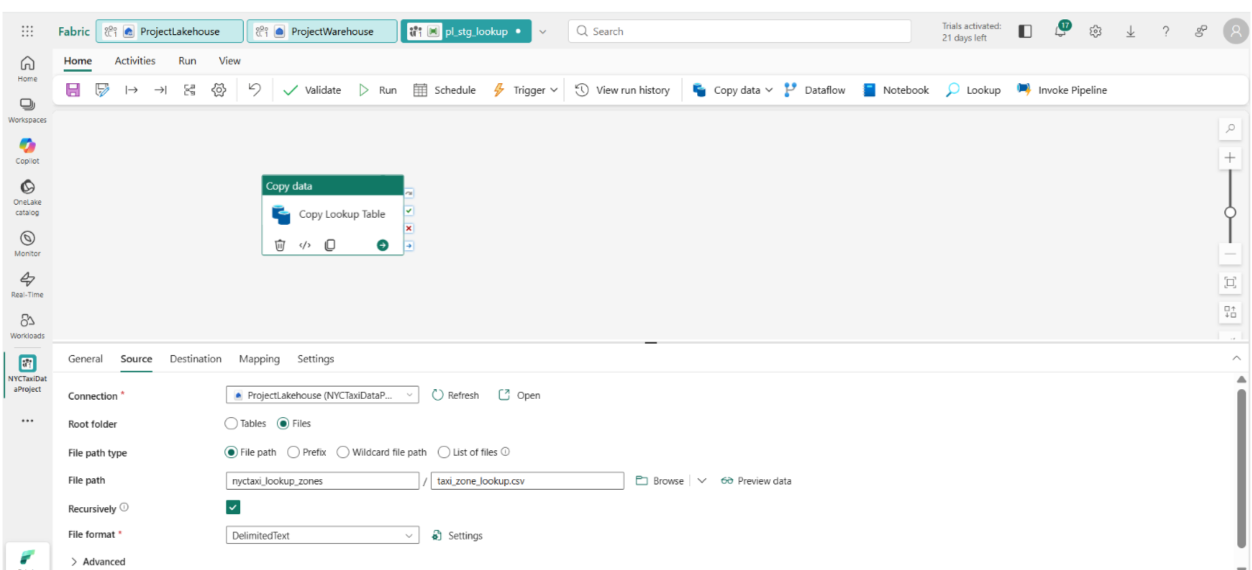
Created a Lakehouse and uploaded the data files in it.



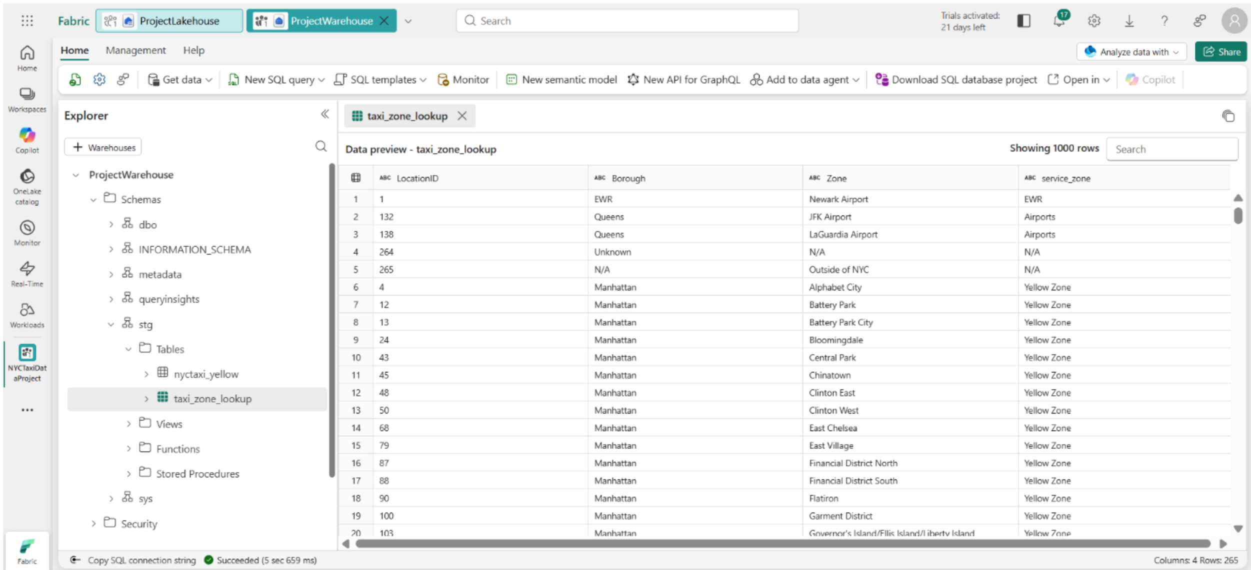


Steps – From Landing to Staging Presentation layer

Created a data pipeline "**pl_stg_lookup**" with copy activity to copy lookup data from lakehouse to staging table in stg schema in the data warehouse.

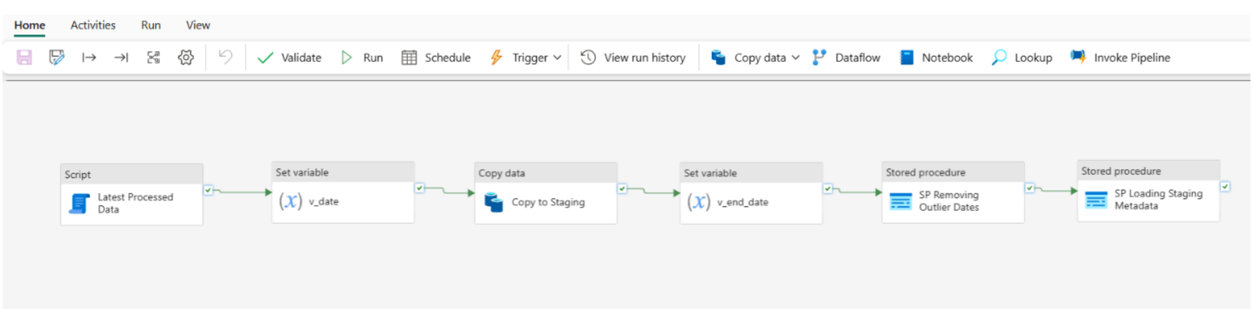


Below is the copied lookup data as it appears in the data warehouse as a staging table under the stg schema after running the pipeline above.



Created pipeline "**pl_stg_processing_nyctaxi**" to ingest taxi trips data from lakehouse to datawarehouse under stg schema.

Code used in **Pipeline: pl_stg_processing_nyctaxi** is as below



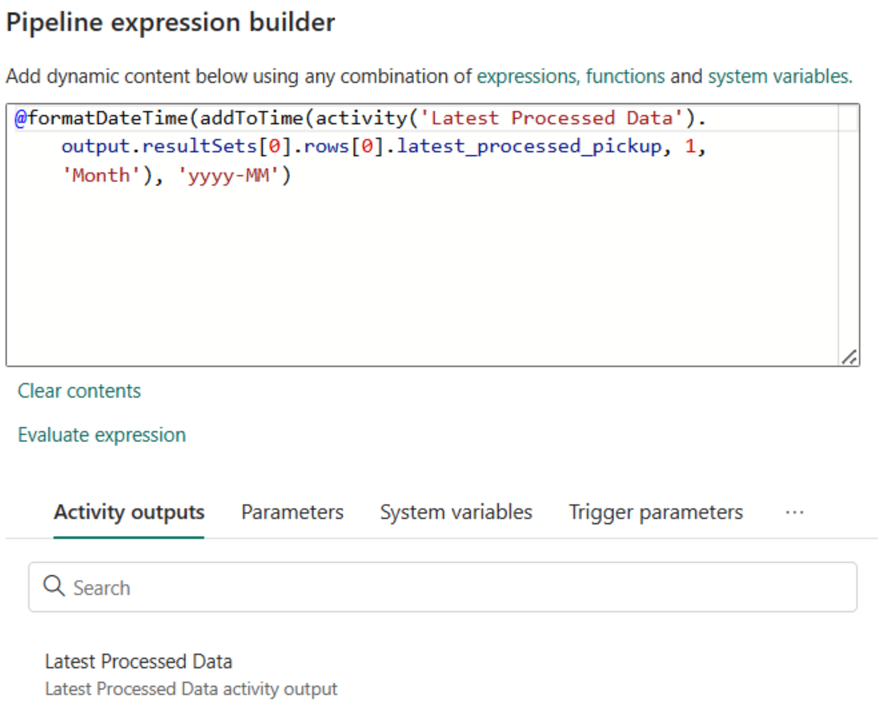
Latest Processed Data

For the Script Activity **“Latest Processed Data”**

```sql
select top 1 latest_processed_pickup

from metadata.processing_log

where table_processed = 'staging_nyctaxi_yellow'

order by latest_processed_pickup desc;
```

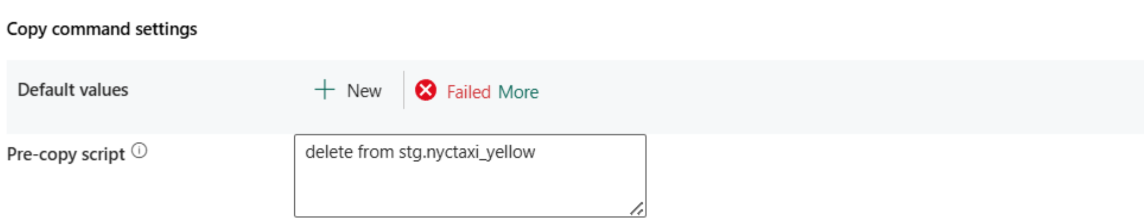
v_date

Pipeline expression for **v_date** Set Variable activity



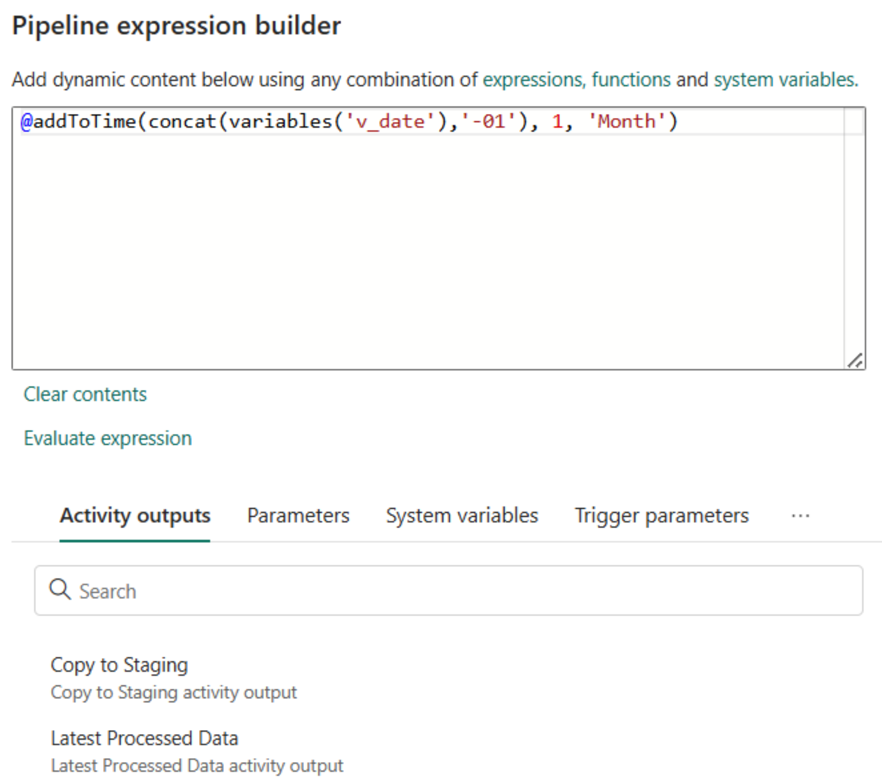
Copy to Staging

Pre Copy Script



v_end_date

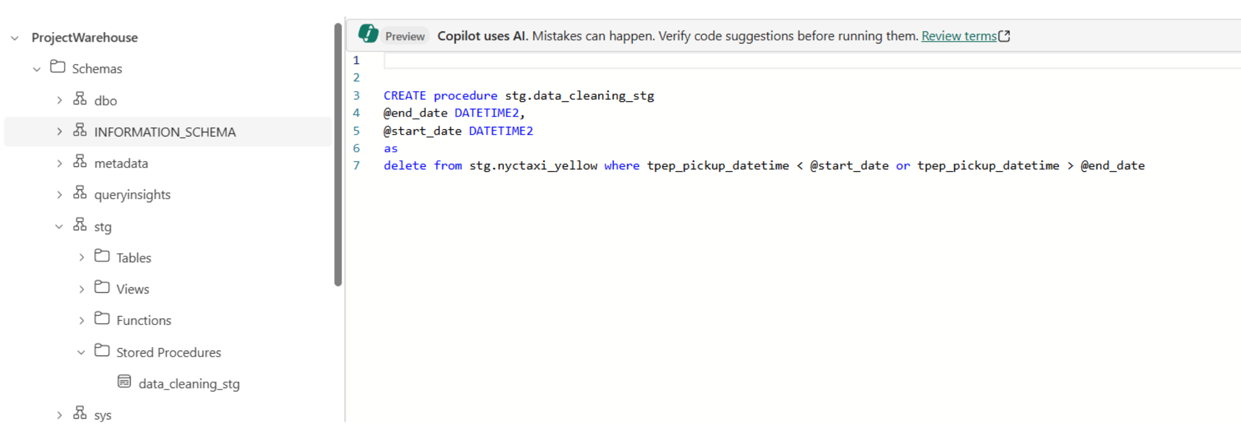
Pipeline expression for **v_end_date** Set Variable activity

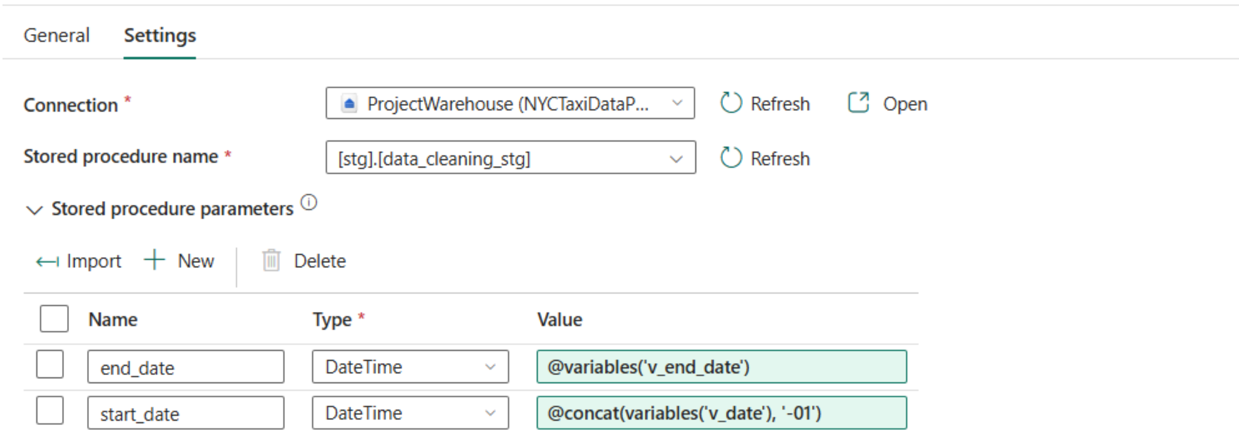


SP Removing Outlier Dates

For the Stored Procedure Activity **“SP Removing Outlier Dates”.**

Create the Stored Procedure **stg.data_cleaning_stg** in the Data Warehouse using the code below.

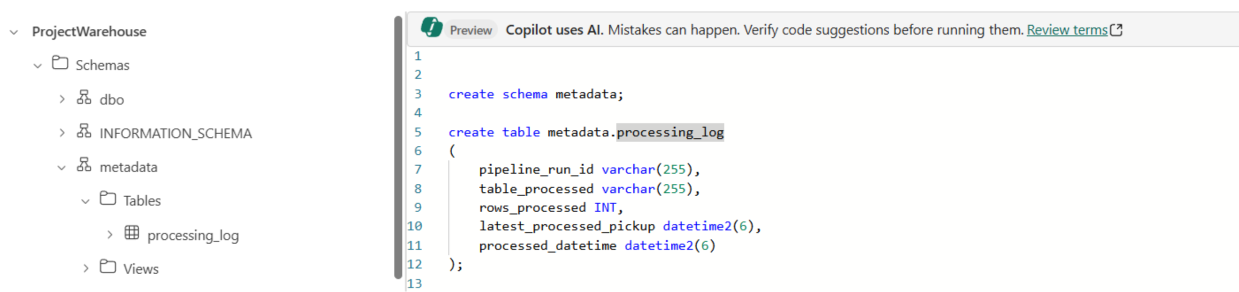




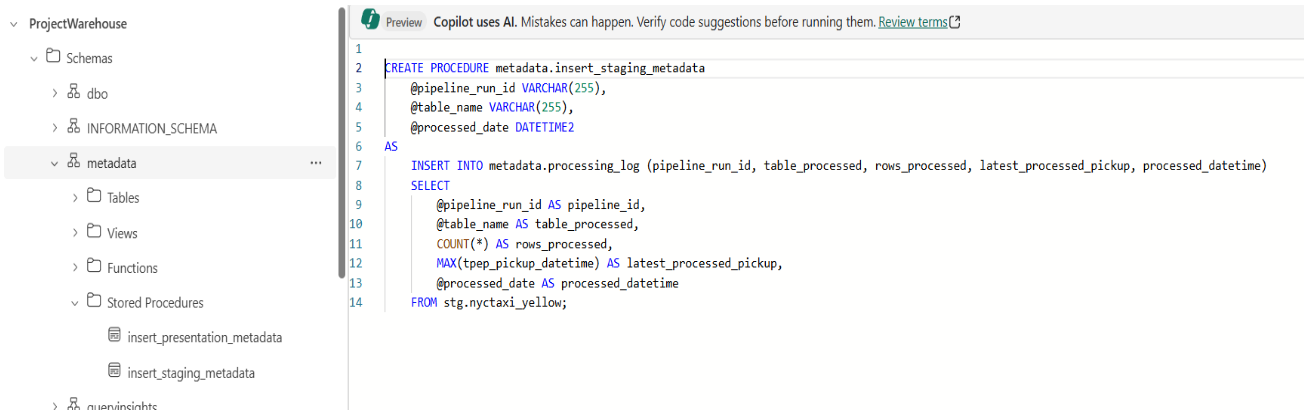
SP Loading Staging Metadata

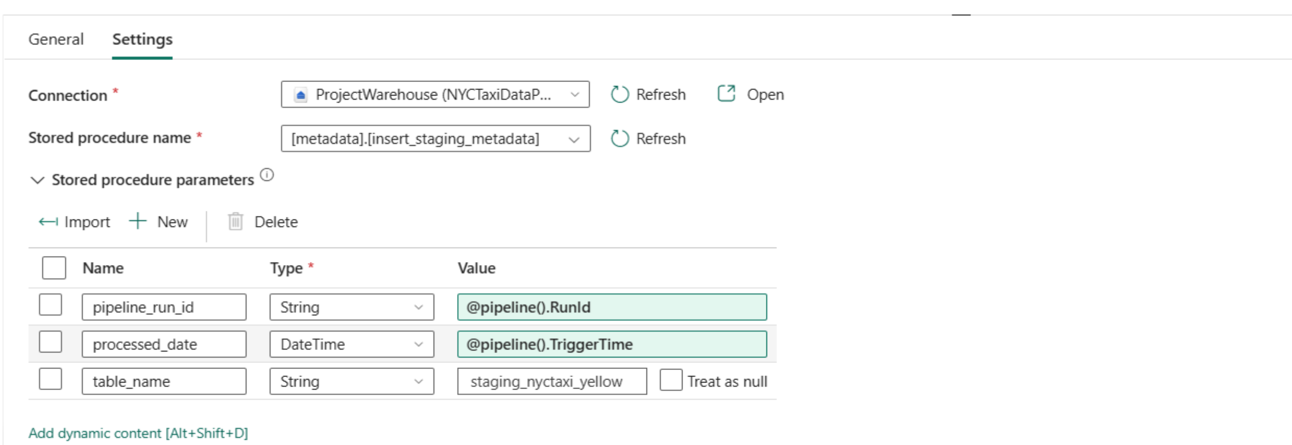
For the Stored Procedure Activity **"SP Loading Staging Metadata".**

Code to create the **metadata.processing_log** table.



Created the Stored Procedure **metadata.insert_staging_metadata** in the Data Warehouse using the code below.



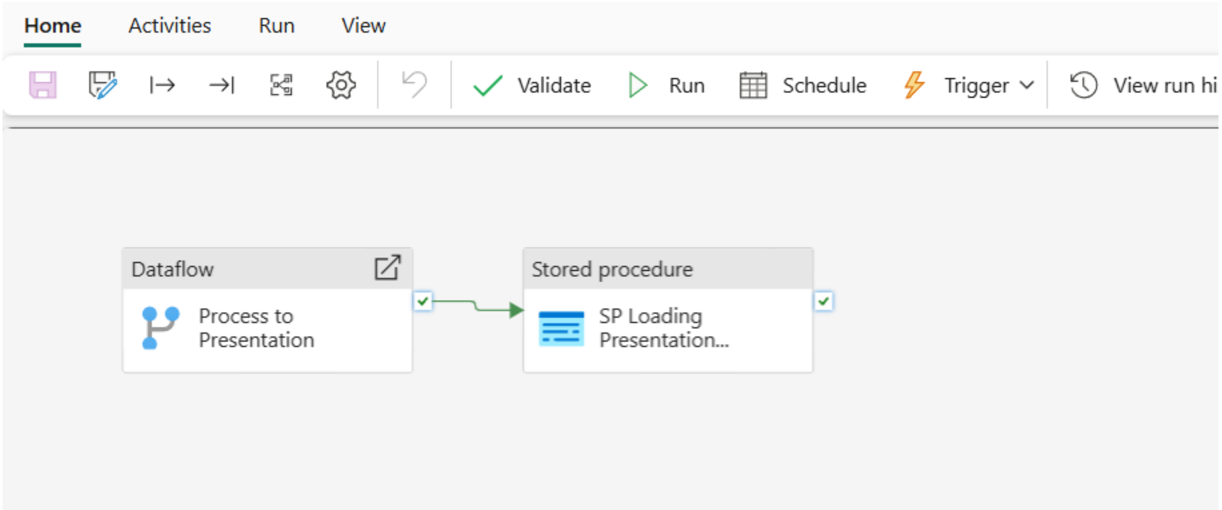


<u>Steps - From Staging to Presentation layer</u>

Overall Pipeline

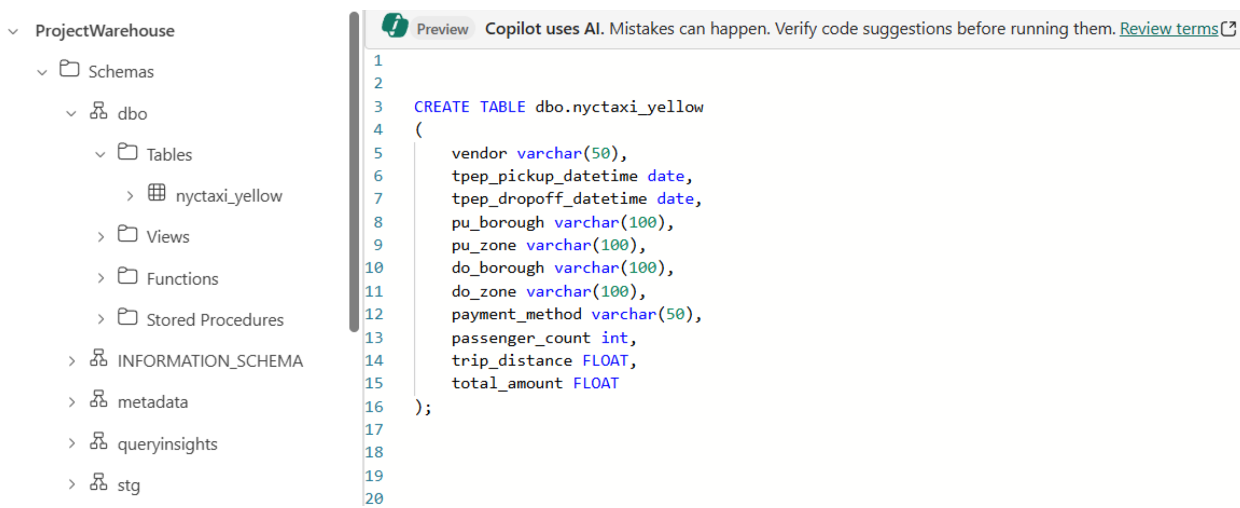
Code used in Pipeline: pl_pres_processing_nyctaxi

This is a view of the pipeline using the Dataflow Gen2 and a Stored Procedure activity.



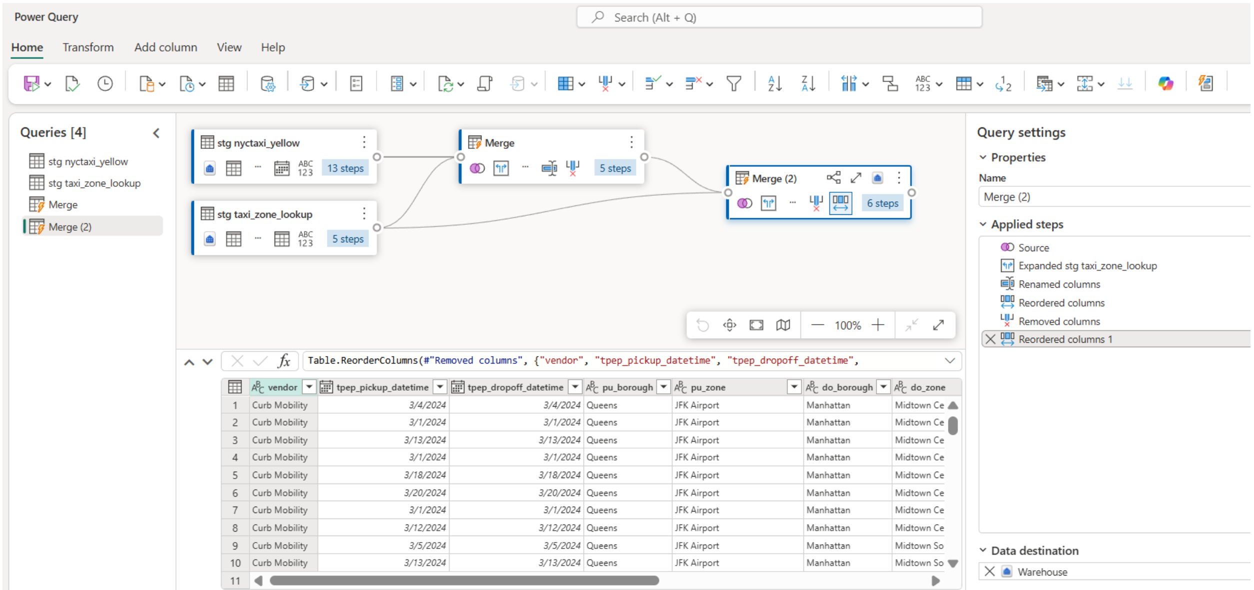
Create the dbo.nyctaxi_yellow table

This is the initial empty table so we can load the data from the Dataflow into this table.



Dataflow - Process to Presentation

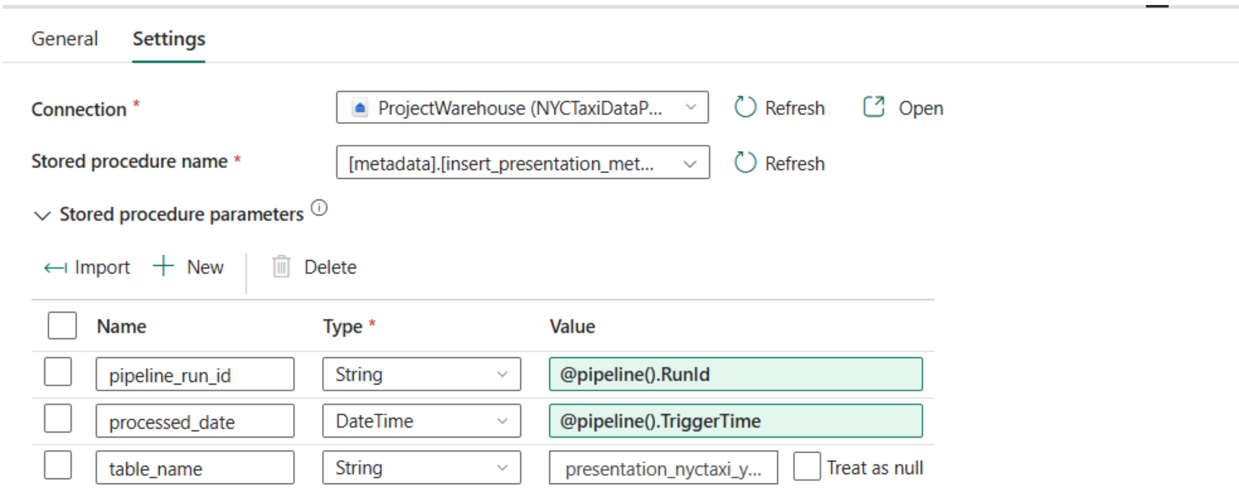
Created the below dataflow to clean, transform and merge the data from the staging tables and load the final data into the destination table **dbo.nyctaxi_yellow table** in the Datawarehouse.



SP Loading Presentation Metadata

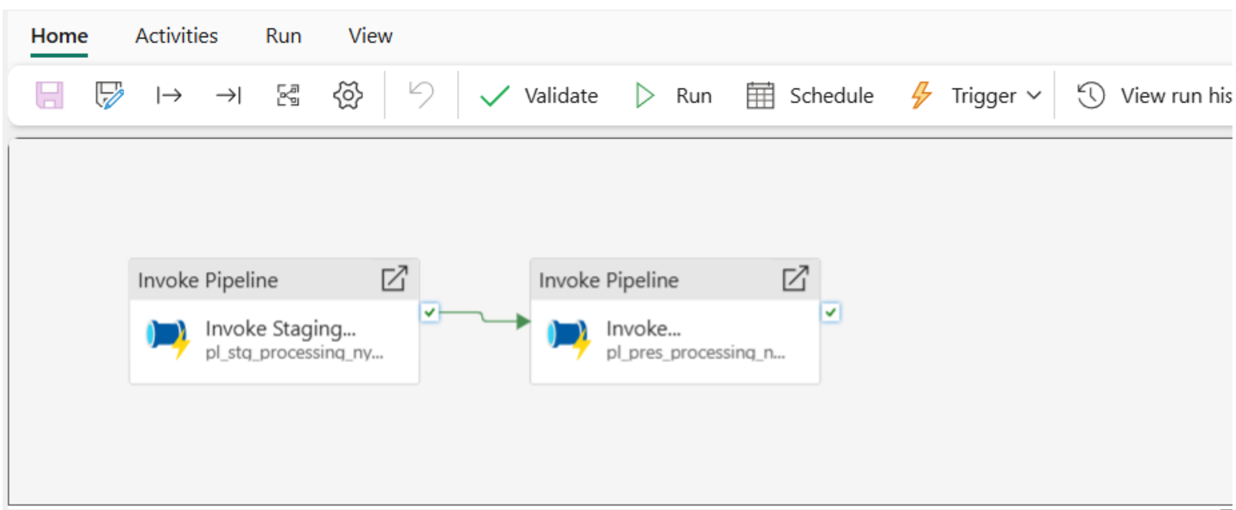
For the Stored Procedure Activity **"SP Loading Presentation Metadata".**

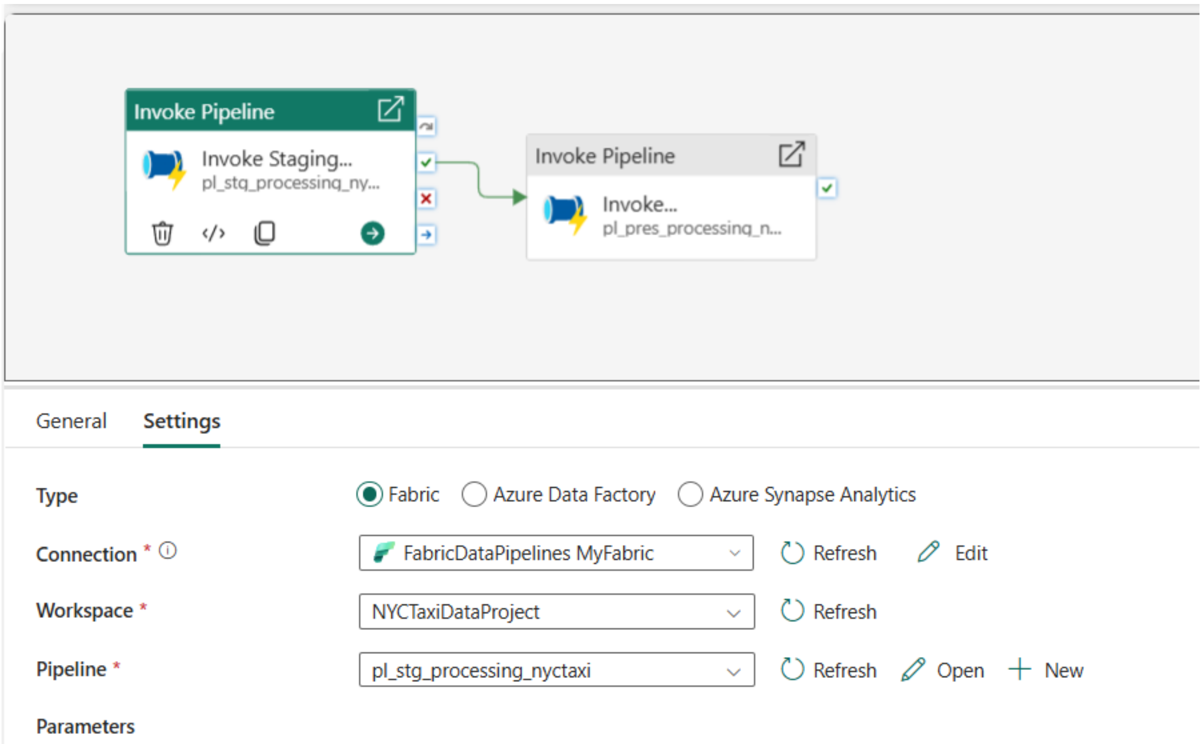
Create the Stored Procedure **metadata.insert_presentation_metadata** in the Data Warehouse using the code below.

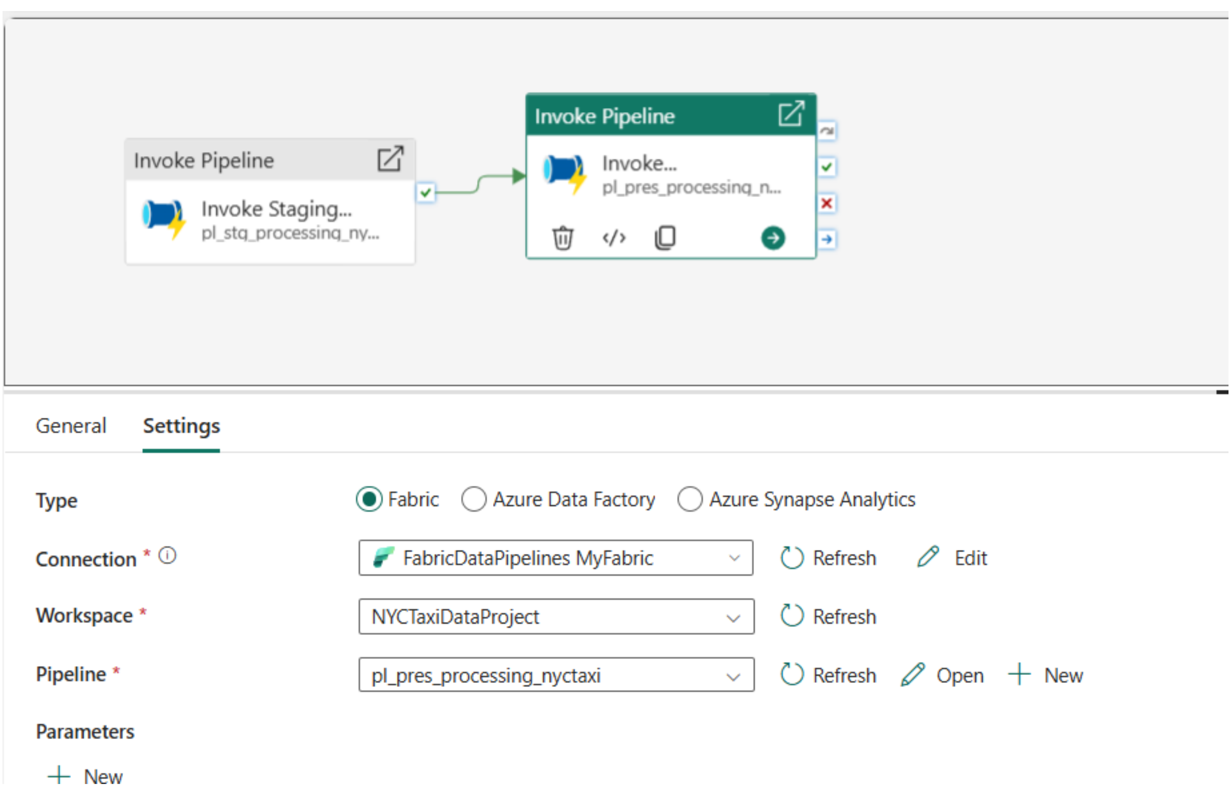


Steps – End-To-End Data Processing

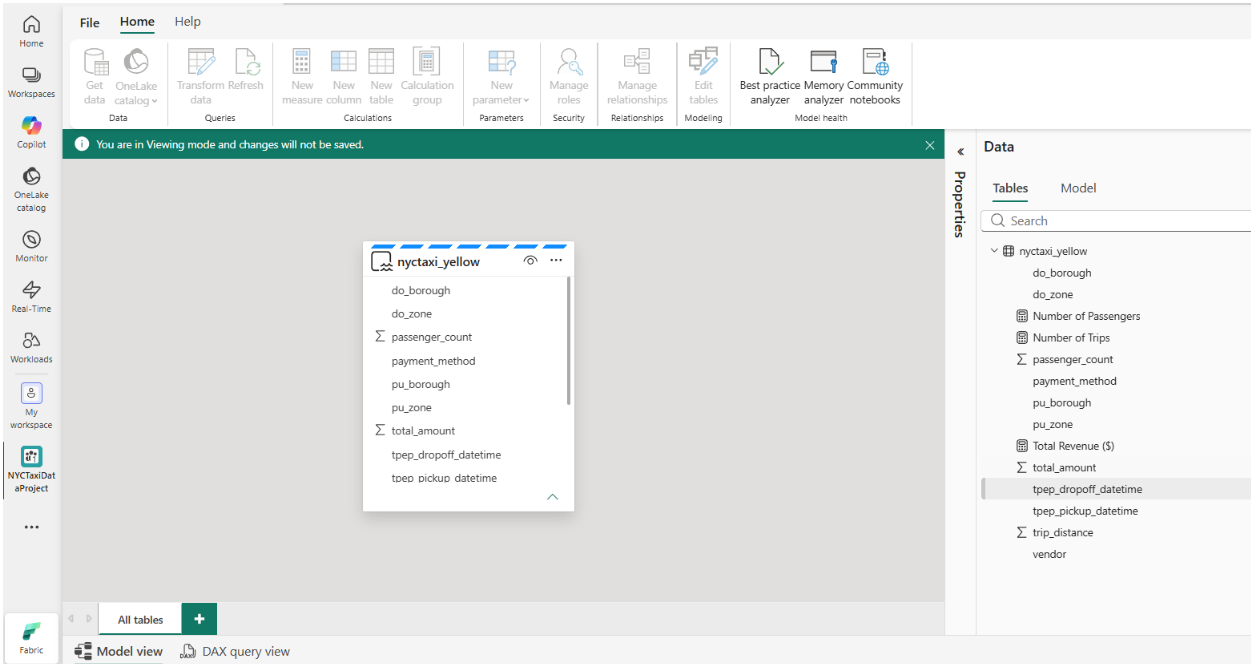
Created an orchestration pipeline "**pl_orchestrate_nyctaxi**" (using "invoke pipeline" activity) that invokes both the pipelines "pl_stg_processing_nyctaxi" and "pl_pres_processing_nyctaxi" that are already created. This orchestration pipeline will run end-to-end data process.




Invoke Pipeline
Invoke Staging...
pl_stg_processing_ny...
Invoke Pipeline
Invoke...
pl_pres_processing_n...
General
Settings
Type
Fabric
Azure Data Factory
Azure Synapse Analytics
Connection *
FabricDataPipelines MyFabric
Refresh
Edit
Workspace *
NYCTaxiDataProject
Refresh
Pipeline *
pl_stg_processing_nyctaxi
Refresh
Open
New
Parameters


Invoke Pipeline
Invoke Staging...
pl_stg_processing_ny...
Invoke Pipeline
Invoke...
pl_pres_processing_n...
General
Settings
Type
Fabric
Azure Data Factory
Azure Synapse Analytics
Connection *
FabricDataPipelines MyFabric
Refresh
Edit
Workspace *
NYCTaxiDataProject
Refresh
Pipeline *
pl_pres_processing_nyctaxi
Refresh
Open
New
Parameters
New

Semantic Model



Power BI Dashboard